

CREDIT CARD CUSTOMER BEHAVIOUR ANALYSIS

1. INTRODUCTION

The aim of this report is to analyse customer credit card data to support early prediction of delinquency risk. By examining behavioural factors such as employment status, credit card type, and monthly payment patterns, the study seeks to identify signals that indicate potential default before it occurs. Through systematic data cleaning and analysis, the dataset is transformed into a reliable foundation for building predictive insights. This enables proactive risk management, helping financial institutions anticipate delinquency and make informed decisions to safeguard both customer relationships and credit portfolios.

2. DATASET OVERVIEW

Total Number of records = 500

Key variables:

SL.NO	COLOUM NAME	DATA TYPE	DISCRPTION
1	Customer_ID	Categorical	Primary Key
2	Age	Numerical	Customer Age
3	Income	Numerical	Annual Income
4	Credit_Score	Numerical	Credit Rating
5	Credit_Utilization	Numerical	Numerical Ratio Of Used Vs. Available Credit
6	Missed_Payments	Numerical	Past Missed Payments
7	Delinquent_Account	Numerical	Past Delinquency
8	Loan_Balance	Numerical	Low Account Balance
9	Debt_to_Income_Ratio	Numerical	Ratio Of Total Debt Payment By Total Income
10	Employment_Status	Categorical	Employed, Self-Employed, Unemployed, Retired, Student
11	Account_Tenure	Numerical	Account Tenure
12	Credit_Card_Type	Categorical	Standard, Gold, Platinum, Business, Student
13	Location	Categorical	Account Holder Location
14	Month_1 to Month_6	Categorical	On-Time, Late, Missed

Notable Missing or Inconsistent Data

1. Income, Credit_Score, Loan_Balance columns have missing value or null values, which may distort financial risk modelling.
2. Employment_Status shows inconsistent formatting (e.g = "EMP", "employed", "Employed"), requiring standardization.
3. Monthly Status Columns (Month_1 to Month_6) have inconsistent labels like "On-time", "On time", and missing entries in some rows.
4. Columns Month_1 to Month_6 are in string format which needs to be in integer for the better analysis of financial model.
5. Credit_Utilization includes values exceeding 1.0, which is unrealistic and likely erroneous.

3. MISSING DATA ANALYSIS

Missing Data Issue	Handling Method	Justification
Income, Credit_Score, Loan_Balance	Mean, Median, Mode imputation by respective score band	Preserves realistic financial behavior while accounting for creditworthiness.
Employment_Status inconsistencies	Standardize categories (e.g., lowercase, unify labels)	Ensures consistent encoding and avoids model confusion due to label fragmentation
Credit_Utilization > 1.0 or missing	Cap at 1.0 and use regression imputation	Prevents unrealistic values and leverages correlated financial features for accuracy.

4. RISK INDICATORS THAT MAY IMPACT DELINQUENCY PREDICTIONS

Indicator	Why It's Important
Missed_Payments ≥ 4	Directly reflects payment behavior and strongly correlates with delinquency.
Credit_Utilization > 0.6	Indicates financial strain and over-reliance on credit, often preceding missed payments.
Debt_to_Income Ratio > 0.4	Suggests limited repayment capacity relative to income, increasing default risk.
Multiple "Missed" or "Late" months	Temporal patterns of poor payment behavior are strong predictors of future delinquency.
Credit_Score < 400	Low scores reflect poor credit history and higher likelihood of default.
Employment_Status = Unemployed/Student	These groups show higher delinquency rates due to unstable or low income.

Based On Above Risk Indicators I Have Made Two Conditions

1. High Risk Customers
 - Condition: Must satisfy 3 or more above condition
 - There are 37 High Risk Customers
2. Medium Risk Customers
 - Condition: Must satisfy 2 above condition
 - There are 151 Medium Risk Customers

Location wise risk customers

Location	High_Risk_Flag	Medium_Risk_Flag
Chicago	8	28
Houston	3	41
Los Angeles	8	29
New York	9	25
Phoenix	9	28

- New York and Phoenix have the highest number of high-risk customers.
- Houston has fewer high-risk customers but the largest group of medium-risk customers.
- Chicago and Los Angeles sit in the middle with moderate levels of both.

5. CONCLUSION & NEXT STEPS

- High Risk (37): These customers are in critical danger of delinquency and need immediate intervention.
- Medium Risk (151): These customers show moderate stress and should be given preventive support before they escalate.
- Low Risk (remaining customers): Do not meet 2 or more conditions, but should still be monitored to prevent migration into higher risk categories.